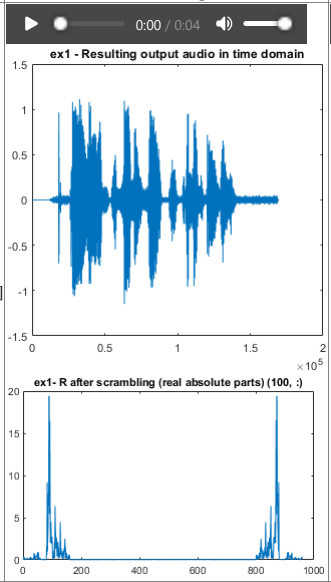
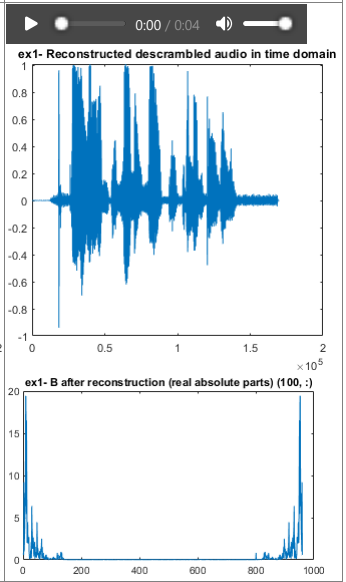
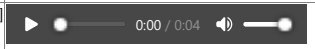
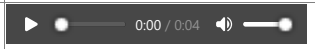
Scrambling Experiment 4

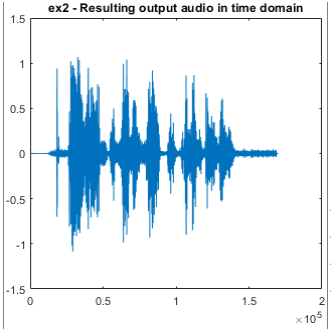
Various Scrambling/Descrambling Configurations

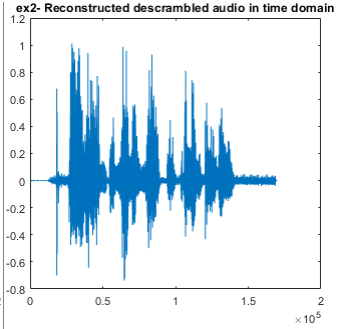
Image resolution for plots: 80

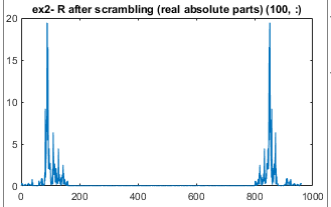
Fs = 48000, Fc = 0.29167

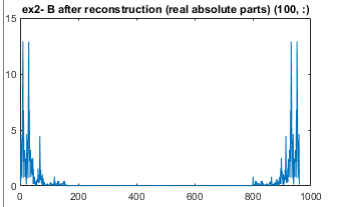
n = 169303, 169920, 169920 for input, scrambled, and reconstructed audio respectively.

Config	Scrambling	Reconstruction
[4 5 6 7 0 1 2 3]		
[4 5 7 6 0 1 2 3]		

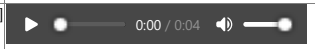

ex2 - Resulting output audio in time domain
$\times 10^5$

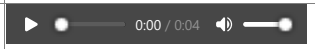

ex2- Reconstructed descrambled audio in time domain
$\times 10^5$

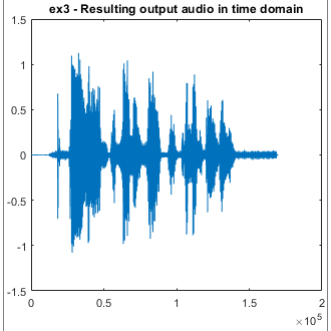

ex2- R after scrambling (real absolute parts) (100, :)

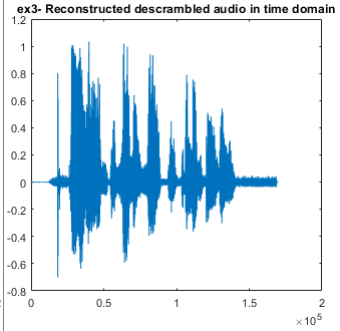

ex2- B after reconstruction (real absolute parts) (100, :)

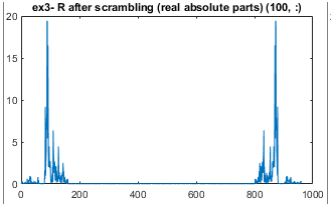
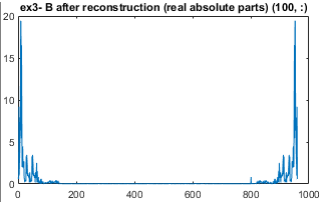
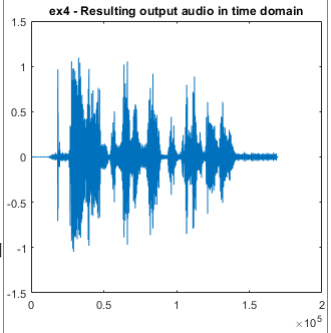
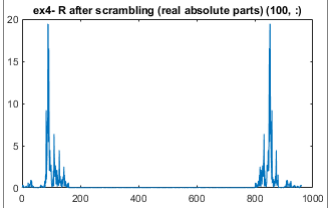
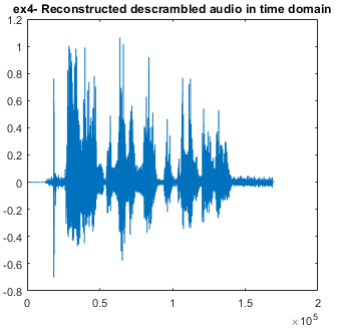
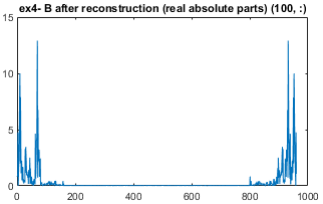
[4 6 5 7 0 1 2 3]

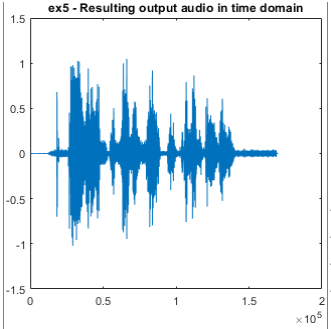

0:00 / 0:04

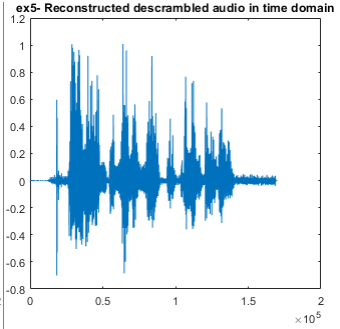

0:00 / 0:04

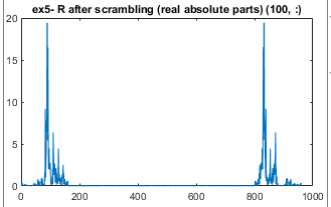

ex3 - Resulting output audio in time domain
$\times 10^5$

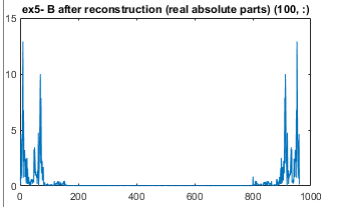

ex3- Reconstructed descrambled audio in time domain
$\times 10^5$

		
[4 6 7 5 0 1 2 3]	0:00 / 0:04  	0:00 / 0:04  
[4 7 5 6 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

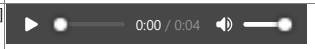

ex5 - Resulting output audio in time domain
$\times 10^5$

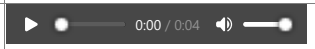

ex5- Reconstructed descrambled audio in time domain
$\times 10^5$

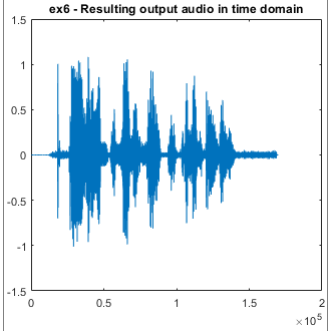

ex5- R after scrambling (real absolute parts) (100, :)

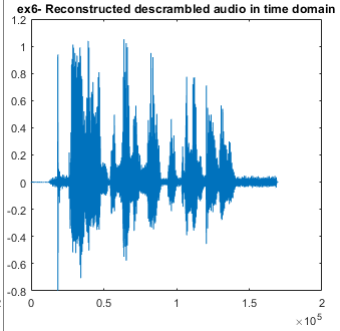

ex5- B after reconstruction (real absolute parts) (100, :)

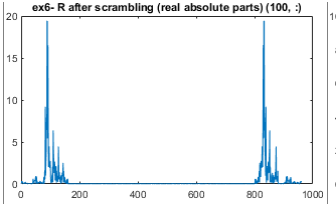
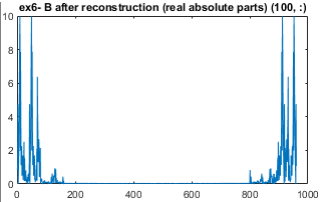
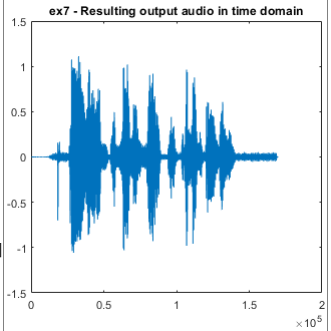
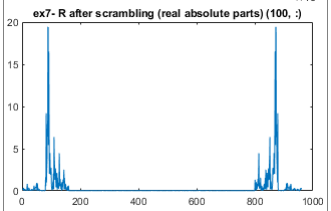
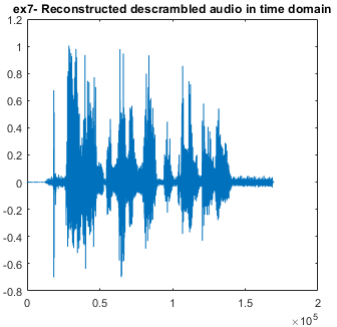
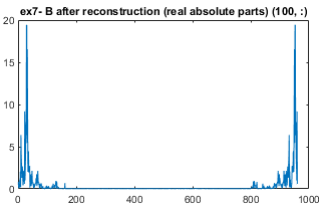
[4 7 6 5 0 1 2 3]

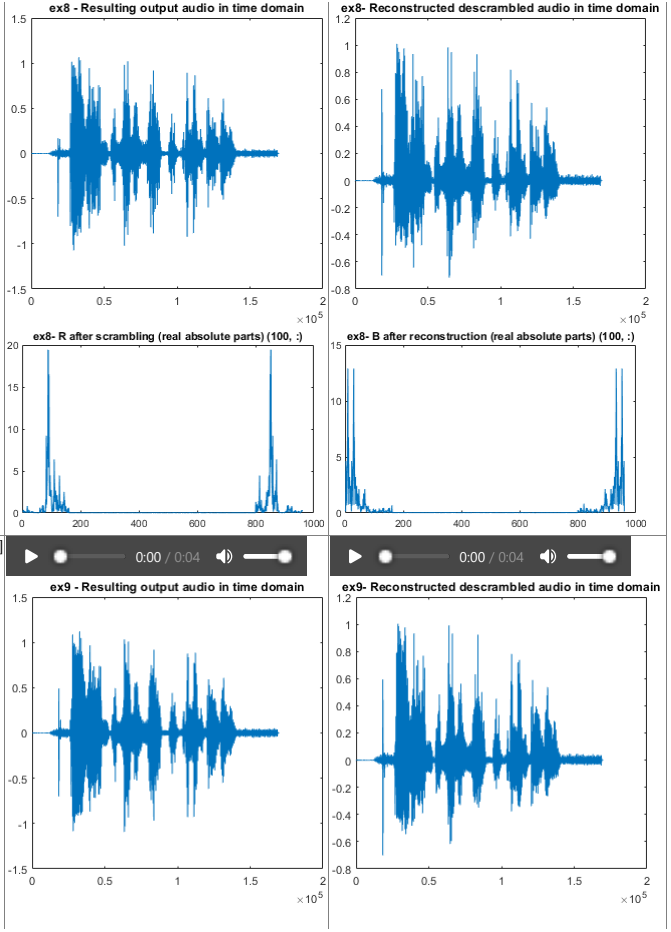

0:00 / 0:04


0:00 / 0:04

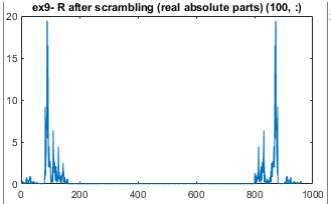
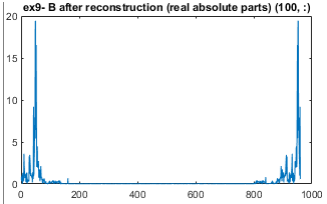
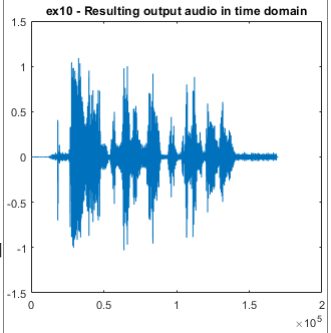
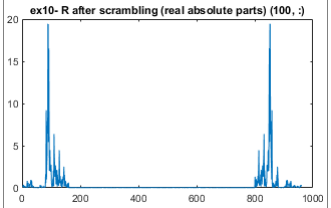
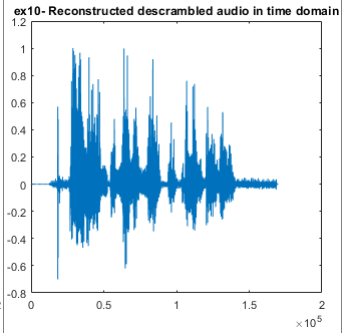
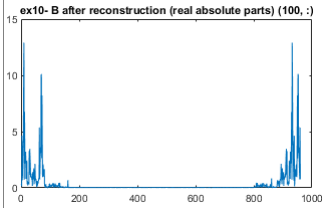

ex6 - Resulting output audio in time domain
$\times 10^5$

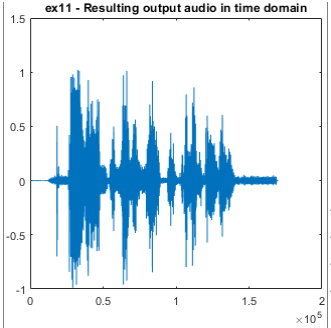

ex6- Reconstructed descrambled audio in time domain
$\times 10^5$

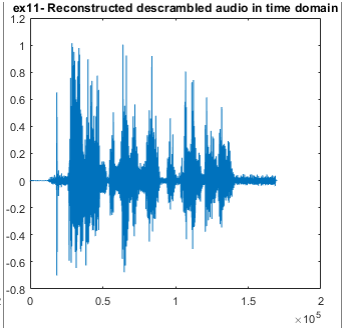
		
[5 4 6 7 0 1 2 3]	0:00 / 0:04  	0:00 / 0:04  
[5 4 7 6 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

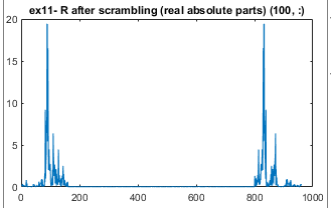

ex8 - Resulting output audio in time domain
ex8- Reconstructed descrambled audio in time domain
ex8- R after scrambling (real absolute parts) (100, :)
ex8- B after reconstruction (real absolute parts) (100, :)
0:00 / 0:04
0:00 / 0:04
ex9 - Resulting output audio in time domain
ex9- Reconstructed descrambled audio in time domain

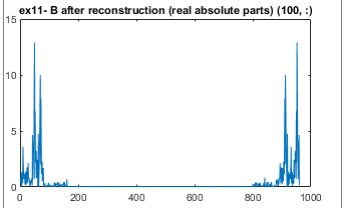
[5 6 4 7 0 1 2 3]

		
[5 6 7 4 0 1 2 3]	0:00 / 0:04  	0:00 / 0:04  
[5 7 4 6 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

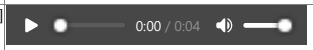

ex11 - Resulting output audio in time domain
1.5
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
×10^5

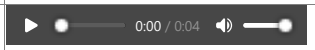

ex11- Reconstructed descrambled audio in time domain
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
0.5
1
1.5
2
×10^5

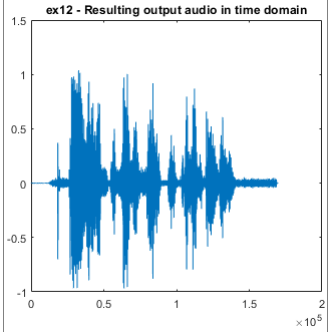

ex11- R after scrambling (real absolute parts) (100, :)
20
15
10
5
0
0
200
400
600
800
1000

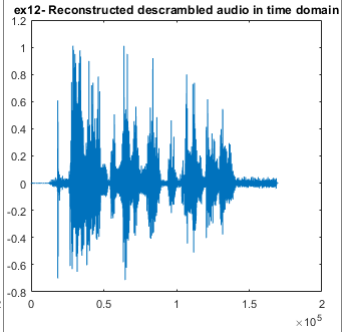

ex11- B after reconstruction (real absolute parts) (100, :)
15
10
5
0
0
200
400
600
800
1000

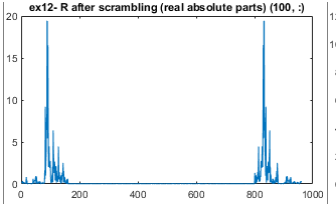
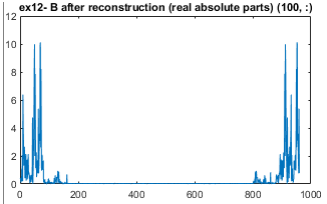
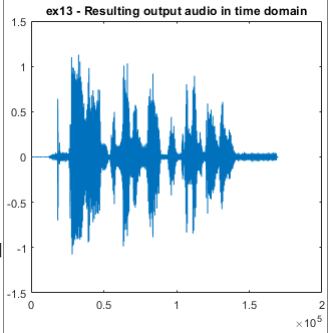
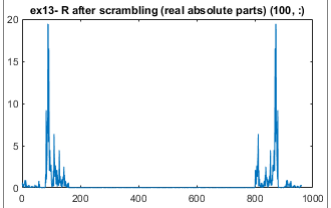
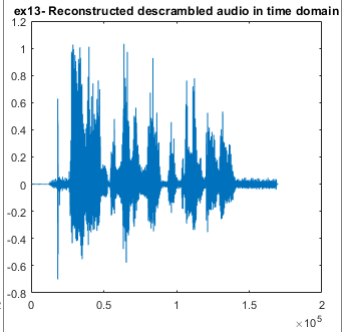
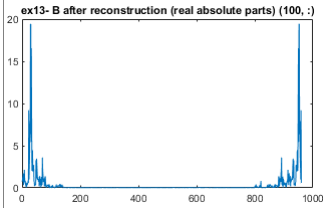
[5 7 6 4 0 1 2 3]

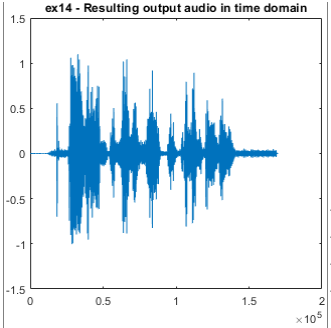

0:00 / 0:04

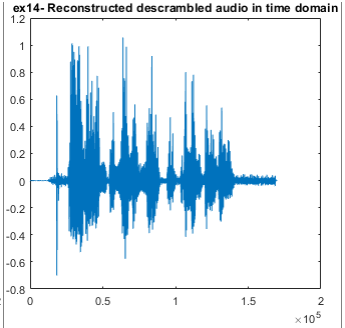

0:00 / 0:04

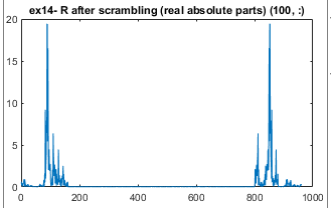

ex12 - Resulting output audio in time domain
1.5
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
×10^5

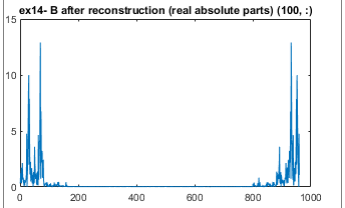

ex12- Reconstructed descrambled audio in time domain
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
0.5
1
1.5
2
×10^5

		
[6 4 5 7 0 1 2 3]	0:00 / 0:04  	0:00 / 0:04  
[6 4 7 5 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

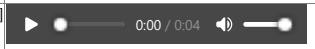

ex14 - Resulting output audio in time domain
$\times 10^5$

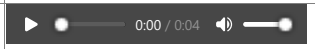

ex14- Reconstructed descrambled audio in time domain
$\times 10^5$

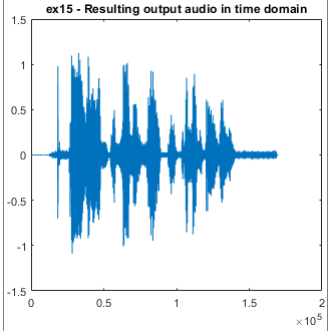

ex14- R after scrambling (real absolute parts) (100, :)

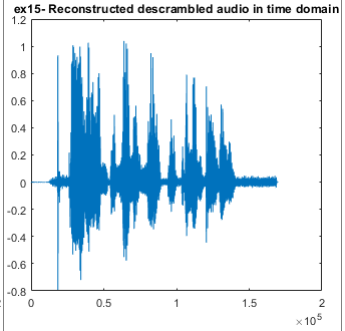

ex14- B after reconstruction (real absolute parts) (100, :)

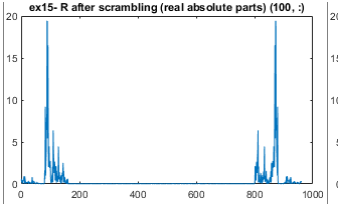
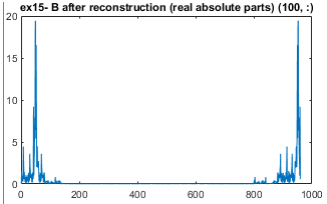
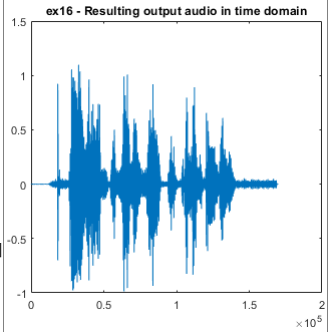
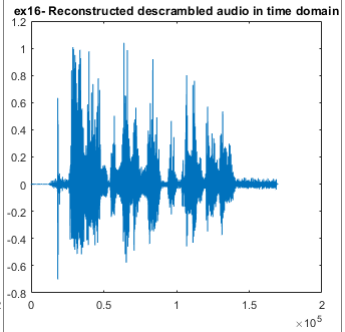
[6 5 4 7 0 1 2 3]

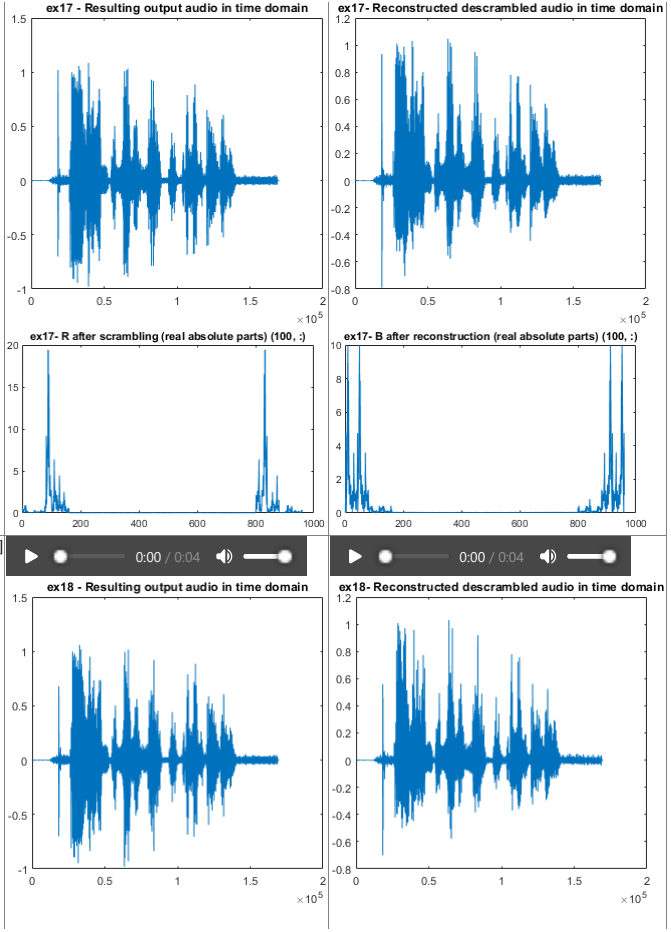

0:00 / 0:04


0:00 / 0:04

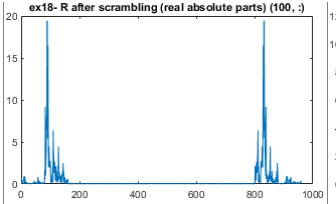
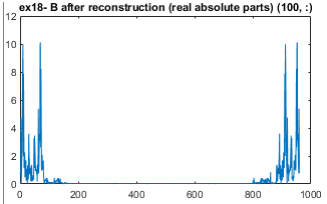
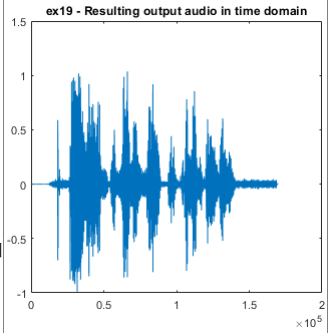
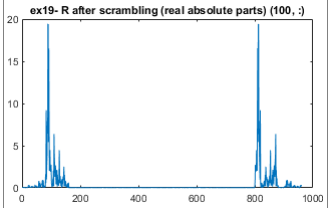
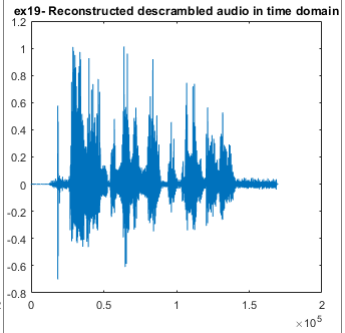
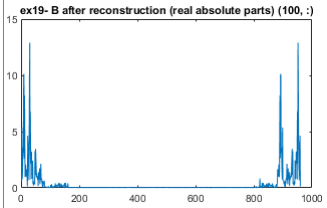

ex15 - Resulting output audio in time domain
$\times 10^5$

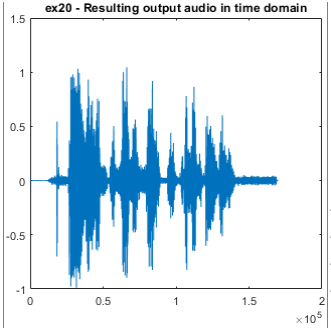

ex15- Reconstructed descrambled audio in time domain
$\times 10^5$

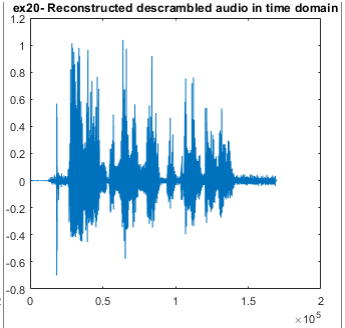
		
[6 5 7 4 0 1 2 3]	0:00 / 0:04 	0:00 / 0:04 
[6 7 4 5 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

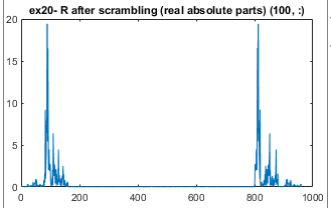

ex17 - Resulting output audio in time domain
ex17- Reconstructed descrambled audio in time domain
ex17- R after scrambling (real absolute parts) (100, :)
ex17- B after reconstruction (real absolute parts) (100, :)
0:00 / 0:04
0:00 / 0:04
ex18 - Resulting output audio in time domain
ex18- Reconstructed descrambled audio in time domain

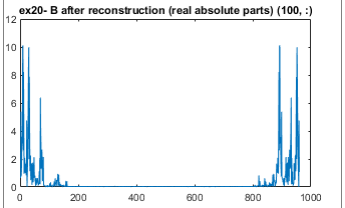
[6 7 5 4 0 1 2 3]

		
[7 4 5 6 0 1 2 3]	0:00 / 0:04  	0:00 / 0:04  
[7 4 6 5 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

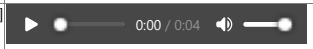

ex20 - Resulting output audio in time domain
1.5
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
×10^5

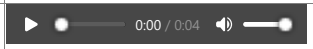

ex20- Reconstructed descrambled audio in time domain
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
0.5
1
1.5
2
×10^5

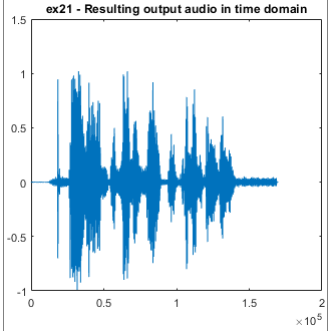

ex20- R after scrambling (real absolute parts) (100, :)
20
15
10
5
0
0
200
400
600
800
1000

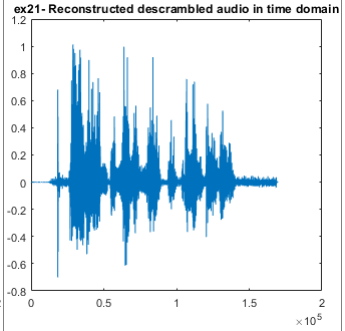

ex20- B after reconstruction (real absolute parts) (100, :)
12
10
8
6
4
2
0
0
200
400
600
800
1000

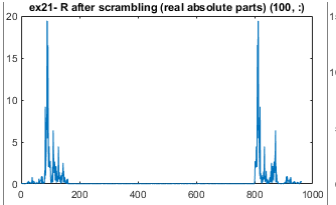
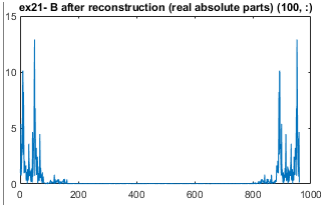
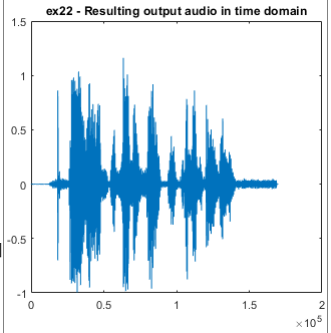
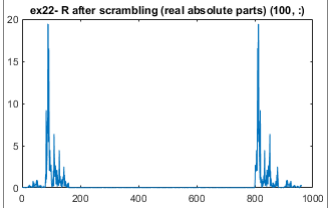
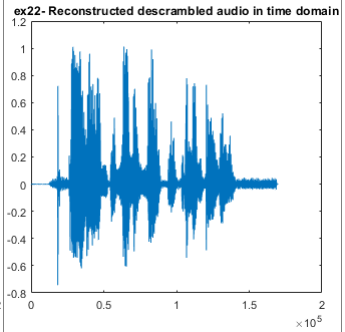
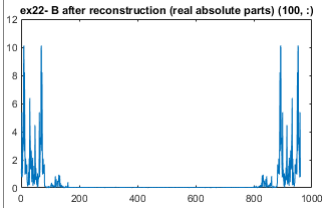
[7 5 4 6 0 1 2 3]

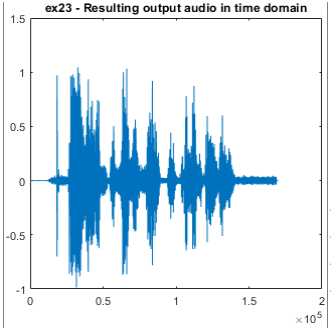

0:00 / 0:04

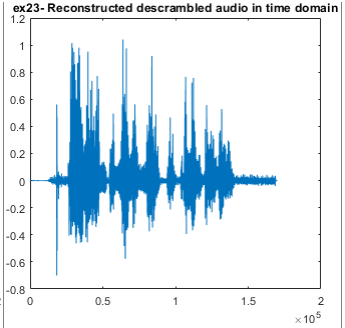

0:00 / 0:04

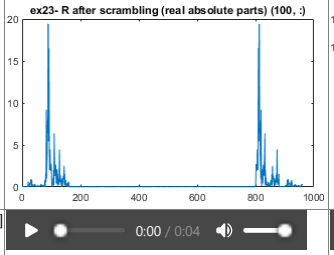

ex21 - Resulting output audio in time domain
1.5
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
×10^5

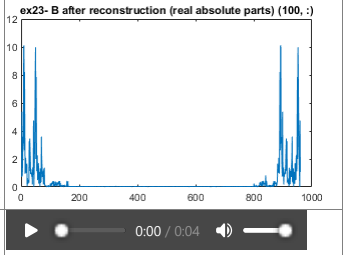

ex21- Reconstructed descrambled audio in time domain
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
0.5
1
1.5
2
×10^5

		
[7 5 6 4 0 1 2 3]	0:00 / 0:04  	0:00 / 0:04  
[7 6 4 5 0 1 2 3]	0:00 / 0:04	0:00 / 0:04


ex23 - Resulting output audio in time domain
1.5
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
×10^5


ex23- Reconstructed descrambled audio in time domain
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
0.5
1
1.5
2
×10^5

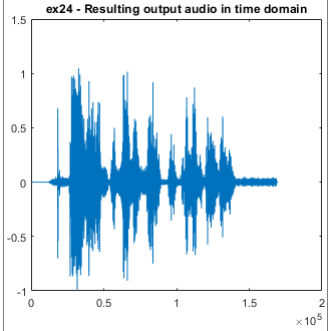

ex23- R after scrambling (real absolute parts) (100, :)
20
15
10
5
0
0
200
400
600
800
1000

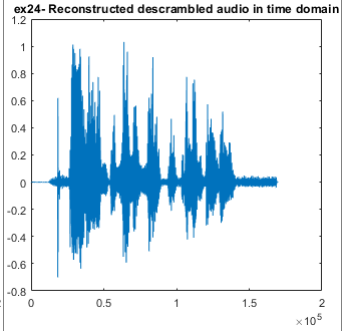

ex23- B after reconstruction (real absolute parts) (100, :)
12
10
8
6
4
2
0
0
200
400
600
800
1000

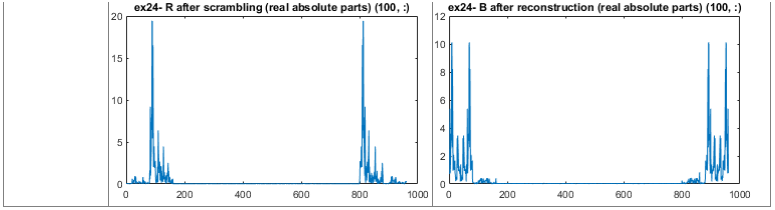
[7 6 5 4 0 1 2 3]

0:00 / 0:04

0:00 / 0:04


ex24 - Resulting output audio in time domain
1.5
1
0.5
0
-0.5
-1
0
0.5
1
1.5
2
×10^5


ex24- Reconstructed descrambled audio in time domain
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
0.5
1
1.5
2
×10^5


ex24- R after scrambling (real absolute parts) (100, :)
20
15
10
5
0
0
200
400
600
800
1000
ex24- B after reconstruction (real absolute parts) (100, :)
12
10
8
6
4
2
0
0
200
400
600
800
1000